

Project Proposal: Hybrid Retrieval on Amazon Reviews Dataset

Abstract

The explosive growth of e-commerce has produced vast volumes of customer feedback in the form of product reviews. The **Amazon Reviews** dataset is a comprehensive collection of hundreds of millions of reviews and rich item metadata that now spans from 1996 to 2023 and includes ratings, review text, helpfulness votes and detailed product attributes

【880832325919793†L8-L28】 【955828709307048†L41-L48】 . These data are invaluable for building information-retrieval systems that help customers discover relevant products or insights. Traditional lexical retrieval methods such as BM25 rely on keyword matching and efficiently provide high recall but cannot capture semantic similarity; purely semantic models based on deep neural networks capture meaning but may miss relevant documents and suffer from high computational cost 【143334345861804†L45-L93】 . Recent research demonstrates that combining lexical and semantic retrieval in a **hybrid system** improves recall and captures complementary evidence 【143334345861804†L96-L106】 . Our project proposes to create a hybrid retrieval system using the Amazon Reviews dataset. We will preprocess a k-core subset of the dataset, build baseline lexical and semantic retrievers, develop algorithms to fuse their results, and evaluate the system using standard information-retrieval metrics. The proposed work aims to demonstrate how hybrid retrieval can enhance the discovery of relevant reviews and support downstream tasks such as recommendation or question-answering.

Problem Definition

Online shoppers often rely on product reviews to make purchasing decisions. However, finding relevant reviews among millions of entries is challenging. Lexical retrieval models like BM25 assign relevance scores based on overlap between query and document terms; they are efficient and provide reasonable recall but struggle with vocabulary mismatch and cannot capture synonyms or paraphrases 【143334345861804†L45-L63】 . Semantic models use deep neural networks to encode queries and documents into dense vectors and measure similarity, allowing them to retrieve documents that use related terminology, but these models alone typically have lower recall and require expensive nearest-neighbour search 【143334345861804†L89-L94】 . A recent study proposed a **lexical-semantic hybrid retrieval approach** where lexical and semantic retrievals run in parallel and their result lists are merged to produce an initial candidate set for reranking 【143334345861804†L96-L106】 . The authors showed that hybrid retrieval can retrieve many relevant documents not covered by lexical approaches alone and improves recall without sacrificing efficiency 【143334345861804†L96-L123】 . We aim to apply this idea to the Amazon Reviews dataset by designing a hybrid retrieval system that uses BM25 as the lexical baseline and transformer-based dense retrieval as the semantic baseline. Key challenges include handling the scale of the dataset, defining meaningful query-document pairs, balancing the contributions of lexical and semantic scores, and evaluating the system with appropriate ground truth.

Dataset

The **Amazon Reviews** dataset curated by Ni et al. and McAuley Lab is one of the largest publicly available collections of e-commerce reviews. The 2018 release contains **233.1 million reviews** with ratings, review text, helpfulness votes and product metadata

【880832325919793†L16-L27】. The newer 2023 release expands the dataset to **571.54 million reviews** (245.2 % larger than earlier versions) and includes finer-grained timestamps, richer item descriptions, and link graphs describing user–item interactions and also-bought/also-viewed relationships 【955828709307048†L54-L66】. Each record includes fields such as:

- **Review content:** textual review, summary title, star rating (1–5), helpfulness votes, and timestamp 【880832325919793†L8-L24】 ;
- **Item metadata:** product description, category, brand, price and image features 【880832325919793†L8-L27】 ;
- **Interaction links:** graphs indicating products that users also viewed or bought 【880832325919793†L12-L27】 【955828709307048†L41-L48】 ;
- **Category splits:** per-category subsets covering dozens of product domains and a 5-core sample where each user and item has at least five reviews 【880832325919793†L96-L100】 .

Because the complete collection is too large for a student project, we will use the **5-core subset** for selected categories (e.g., Electronics, Home & Kitchen) or the “ratings-only” files to build a manageable corpus. We will construct query–document pairs by using product titles or user questions as queries and concatenating review text and summaries as documents. The also-bought/also-viewed links and item categories will serve as ground-truth relevance labels for evaluation.

State-of-the-Art Methods & Algorithms

1. **Lexical retrieval (BM25)** – Traditional information-retrieval systems rank documents by counting the frequency and rarity of query terms. BM25 is widely used as the baseline because it assigns relevance scores based on term frequency and inverse document frequency and is efficient due to its inverted-index implementation 【143334345861804†L45-L53】. However, BM25 cannot match synonyms or capture semantic context and struggles when relevant documents do not share surface terms 【143334345861804†L55-L73】 .
2. **Semantic retrieval (Dense retrieval using Transformers)** – Semantic retrieval encodes queries and documents into continuous vector spaces using transformer models such as Sentence-BERT. By computing the cosine similarity between vector embeddings, semantic retrieval can identify documents that are semantically related to a query even when they have few exact term overlaps 【143334345861804†L75-L84】 . Dense retrieval requires approximate nearest-neighbour (ANN) indexing (e.g., FAISS, ScaNN) to achieve acceptable latency.
3. **Hybrid retrieval** – A hybrid approach combines lexical and semantic retrieval by running both models in parallel and merging or scoring the results. Kuzi et al. showed

that the hybrid approach can improve recall because the semantic model retrieves relevant documents not captured by lexical matching 【143334345861804†L96-L100】 . They propose to interleave the top-k results from both models or to combine scores with a weighted scheme, and demonstrate that this method is efficient and does not require access to large-scale query logs 【143334345861804†L96-L123】 .

4. **Learning-to-rank and reranking models** – Beyond retrieval, reranking models such as cross-encoders or LambdaMART can reorder the combined candidate list to optimize metrics like normalized discounted cumulative gain (nDCG). These models take query–document pairs and compute a relevance score using deep attention mechanisms. They can improve precision at top ranks but are computationally expensive, so they operate on the reduced candidate set produced by the hybrid retriever.

Research Plan

Our research will proceed through several stages:

1. **Data acquisition and preprocessing** – Download the 5-core Amazon Reviews subsets for chosen categories (e.g., Electronics, Books). Parse the JSON data to extract review text, summary titles, star ratings and item metadata. Build dictionaries of product titles and categories. Clean the text by removing HTML tags, lower-casing, tokenizing and filtering stop words. Construct ground-truth relevance by using the also-bought/also-viewed graphs and by assuming that reviews of the same item are relevant to the item’s title.
2. **Baseline lexical retrieval** – Implement a BM25 retrieval engine using an open-source IR toolkit (e.g., Anserini or Pyserini). Index the review documents and evaluate retrieval for a set of queries using recall@K and mean average precision. Analyse strengths and weaknesses by inspecting cases where BM25 fails to retrieve semantically relevant reviews 【143334345861804†L55-L74】 .
3. **Baseline semantic retrieval** – Fine-tune a pretrained Sentence-BERT or MiniLM model on weakly supervised pairs derived from the dataset, such as (item title, review text). Use approximate nearest-neighbour search (FAISS) to index the vector embeddings of documents and perform dense retrieval. Measure recall and precision, noting that semantic retrieval may capture synonyms but could miss documents due to lower recall 【143334345861804†L89-L100】 .
4. **Hybrid retrieval implementation** – Develop methods to combine lexical and semantic retrieval results. We will experiment with: (a) interleaving top-k lists from both models; (b) linear combinations of normalized BM25 and semantic scores; and (c) rank fusion techniques such as Reciprocal Rank Fusion. We will also explore weighting parameters using a held-out validation set.
5. **Reranking and ranking models** – Train a learning-to-rank model using a small labelled set of query–document pairs (e.g., queries from user questions answered by reviews). Use cross-encoders (BERT-based) or boosted decision trees (LambdaMART) to rerank the top candidates from the hybrid retriever, optimizing nDCG and precision at K.

6. **Evaluation** – Evaluate each retrieval method on a held-out set of queries with ground-truth relevant documents. Use standard IR metrics: precision at K (P@K), recall at K (R@K), mean reciprocal rank (MRR), mean average precision (MAP), and normalized discounted cumulative gain (nDCG). MAP averages the precision of each query across the ranked list [454040562745790†L461-L467] , while nDCG measures the usefulness of documents based on position and graded relevance [454040562745790†L475-L499] . We will also assess latency and memory usage to understand efficiency trade-offs. Qualitative analysis will include case studies illustrating how lexical and semantic retrieval complement each other.
7. **Ablation and analysis** – Investigate how different model sizes, embedding dimensions and fusion strategies affect performance. Test the sensitivity to the number of retrieved candidates. Analyse failure cases and the effect of class imbalance (e.g., items with few reviews). Document findings and propose future improvements.

Evaluation Plan

The effectiveness of our hybrid retrieval system will be assessed using offline test collections. The evaluation plan will consider the following metrics and aspects:

- **Recall@K and Precision@K** – Determine the proportion of relevant documents retrieved at rank K and the proportion of retrieved documents that are relevant. These metrics directly reflect the ability of lexical, semantic and hybrid models to retrieve relevant reviews.
- **Mean Reciprocal Rank (MRR)** – Evaluate the position of the first relevant document in the ranking; important for question-answering scenarios.
- **Mean Average Precision (MAP)** – Compute the mean of the average precision scores across queries, which summarises the quality of the entire ranked list [454040562745790†L461-L467] .
- **Normalized Discounted Cumulative Gain (nDCG)** – Apply graded relevance judgements and penalise relevant documents appearing lower in the list by computing DCG and normalizing it with the ideal ranking [454040562745790†L475-L499] .
- **Latency and Efficiency** – Measure query latency and index size for the lexical, semantic and hybrid retrieval pipelines to ensure the system is deployable.
- **Robustness and Ablation Studies** – Evaluate the effect of dataset sampling, model size and fusion strategies on performance. Perform cross-category evaluation to test generalization. Use statistical significance tests (e.g., paired t-test) to confirm improvements.

Project Timeline & Division of Work

Team Members	Task	Proposed Deadline
Member A & Member B	Acquire and preprocess the Amazon Reviews 5-core	10/10/25

Team Members	Task	Proposed Deadline
	subsets; parse reviews and metadata; construct queries and ground-truth relevance sets; address class imbalance and sampling issues.	
Member B & Member C	Implement and tune the BM25 lexical retrieval baseline using an IR toolkit; evaluate and analyse retrieval effectiveness .	10/20/25
Member C & Member D	Fine-tune a transformer- -based semantic retrieval model on weakly supervised pairs; index embeddings with FAISS; benchmark dense retrieval performance .	10/30/25
Member A & Member D	Develop	11/05/25

Team Members	Task	Proposed Deadline
	hybrid retrieval strategies (list interleaving, score fusion, reciprocal rank fusion); tune fusion weights on validation data.	
Member B & Member D	Train reranking models (cross-encoder or LambdaMA RT) on labelled data; evaluate the impact on nDCG and precision at top ranks.	11/10/25
All Members	Conduct comprehensive evaluation and ablation studies; analyse results; prepare final report and presentation.	11/15/25

Note: Team member names can be updated to match the actual project participants.

References

1. Ni, Jianmo; Li, Jiacheng; McAuley, Julian. **Amazon review data (2018)**. The dataset includes ratings, review text, helpfulness votes and product metadata and expands to 233.1 million reviews 【880832325919793†L8-L27】 .
2. McAuley Lab. **Amazon Reviews'23**. The 2023 release contains 571.54 million reviews, richer metadata and user-item graphs 【955828709307048†L54-L66】
【955828709307048†L41-L48】 .
3. Kuzi, Saar et al. **Leveraging Semantic and Lexical Matching to Improve the Recall of Document Retrieval Systems: A Hybrid Approach**. The paper proposes running lexical and semantic retrieval in parallel and merging the results to improve recall 【143334345861804†L96-L106】 ; it shows that lexical models like BM25 rely on term overlap 【143334345861804†L45-L53】 and that hybrid retrieval retrieves relevant documents missed by lexical models 【143334345861804†L96-L100】 .
4. **Evaluation measures (information retrieval)**. Wikipedia article on IR metrics. Mean average precision (MAP) is defined as the mean of the average precision scores across queries 【454040562745790†L461-L467】 . Normalized discounted cumulative gain (nDCG) is computed by dividing DCG by the ideal DCG and measures the usefulness of documents based on their rank 【454040562745790†L475-L499】 .